

What a Doppler Weather Radar measures

- Reflectivity: how much energy comes back (dBZ)
- Radial velocity: motion toward or away from the radar
- Spectrum width: turbulence within the beam

The volume scan strategy

- Antenna turns 360° at a series of elevation angles
- Lowest elevations see closest to the ground
- One full volume every few minutes

Pre-shift checks

- Transmitter, antenna and servo status
- Open alarms and the previous shift's notes
- Data reaching the central server on time

Reading reflectivity

- Around 20 dBZ: light rain
- 40 dBZ and above: moderate to heavy rain
- 50 dBZ and above: very heavy rain, possible hail

Reading radial velocity and aliasing

- Colours show motion toward and away from the radar
- Nyquist velocity sets the largest speed measured cleanly
- Aliasing: a sudden jump from strong inbound to strong outbound

Handling routine alarms

- Sort alarms: information or service-affecting
- Follow the station SOP step by step
- Log the time and escalate to engineering

Handing over a shift safely

- Radar status and open alarms
- Current weather and products issued
- Pending tasks, signed by both officers